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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2024

Code: 101403 ***Time: 03 Hours***

Semester: IV ***Subject: Engineering Geology*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option; write the answer of the following (Any seven): **[2 x 7 = 14]**

(a) The property of a mineral by virtue of which it can be cut with a knife is
(i) Parting (iii) Malleable
(ii) Sectile (iv) Ductile

(b) When the mineral occurs in flattened or square form, it is called
(i) Tabular (iii) Botroidal
(ii) Elongated (iv) Box

(c) Kyanite shows which form?
(i) Columnar (iii) Bladed
(ii) Elongated (iv) Tabular

(d) The arrangement made near top of dam for passage of excess water from the reservoir is
(i) Sluice way (iii) Gallery
(ii) Spillway (iv) Abutments

(e) Chromite shows which type of structure?
(i) Granular (iii) Reniform
(ii) Globular (iv) Mammillary

(f) A temporary dam constructed to exclude water from a specific area is
(i) Debris dam (iii) Cofferdam
(ii) Diversion dam (iv) Storage dam

(g) What occurs due to fall of big rock blocks due to release of stresses during tunneling?
(i) Rock fall (iii) Blockage
(ii) Rock burst (iv) Water rush

(h) Mica is
(i) Flexible (iii) Flexible and elastic
(ii) Rigid (iv) Elastic

(i) Granite is which type of rock?
(i) Igneous Rock (iii) Metamorphic Rock
(ii) Sedimentary Rock (iv) All of these

(j) Hardness of human finger nail is
(i) 1.5 (iii) 3
(ii) 2 (iv) None of these

Q.2 Write the answers in short: **[7 x 2 = 14]**

(a) What is Rock forming mineral? Discuss the process of formation of mineral in nature.

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(b) Give a Detailed account on the physical properties of feldspar.

Q.3 Write the answers in short: **[7 x 2 = 14]**

(a) Write the engineering properties and uses of: (i) Granite (ii) Dolerite (iii) Basalt (iv) Sandstone

(b) Distinguish between Igneous and sedimentary rocks.

Q.4 Write the answers in short: **[7 x 2 = 14]**

(a) Enumerate the mitigation measures for Slope instability.

(b) What are the factors that influence slope instability?

Q.5 Write the answers in short: **[7 + 5 = 12]**

(a) Illustrate the various types of folds with neat sketches.

(b) Write a note on seismic zone in India.

Q.6 Write the answers in short: **[7 x 2 = 14]**

(a) Illustrate the various types of fault with neat sketches.

(b) Critically discuss the concept of plate tectonics and its role in generation of Earthquake.

Q.7 Write the answers in short: **[7 x 2 = 14]**

(a) Discuss briefly on Rock Bolting, Rock Anchoring and Retaining Wall.

(b) Discuss the creation of reservoir. Draw a sketch to show a dam with reservoir and label it.

Q.8 Write the answers in short: **[7 x 2 = 14]**

(a) What is Dam and its Utility? Draw the sectional view of a dam.

(b) Write an essay on geological condition required for design and construction of dam.

Q.9 Answer any two of the following: **[7 x 2 = 14]**

(a) Explain the geological causes for the failure of the Dam.

(b) Explain the suitability of different Rocks in Dam construction.

(c) Enumerate with appropriate figures on the types, causes of landslides and their mitigation measures.

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